

Where Nobody Lives

Building a national map of the census blocks that hold no people

Democracy's Data

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The census block is the smallest unit the Census Bureau publishes, and there are 8,174,955 of them. 2,368,443 hold nobody at all: 29.0% of the blocks in the country recorded a population of zero in 2020. That is not a defect in the count. It is what happens when you cut a country into pieces small enough that most pieces are a lake, an interchange, a runway, a cemetery, or a stretch of national forest.

This brief is a walkthrough. It shows where each file came from, what was done to it, and what the resulting map does and does not say. The point is not the map at the end; it is that you could rebuild it.

01 Where the data came from

Two files, from two different parts of the Bureau, joined on one key.

The population. Every block's 2020 count comes from the Census Data API, from the 2020 Decennial Census P.L. 94-171 Redistricting Data Summary File, at <https://api.census.gov/data/2020/dec/p1>. The variable is P1_001N, total population. A request needs a free key, and the API will not return blocks for a whole state at once, so `build-block-pop.R` walks the country one county at a time, roughly 3,100 requests.

Three properties of this file are why it is the right one. It is a complete count, not a sample, so there are no margins of error. Its atom is the block, so nothing smaller is being hidden. And it is the file state legislatures actually receive for redistricting, which makes it the operative number rather than an academic one.

The shapes. Block boundaries come from TIGER/Line 2020, the Bureau's digital outlines, one zipped shapefile per state, about 10 GB for the country. State outlines for the map's frame come from the Bureau's generalized cartographic boundary files, which are drawn for mapping rather than for measurement and are clipped to the shoreline.

A detail worth stealing: the TIGER block file already carries a POP20 field, so the map could have been built from the shapefiles alone and the API skipped entirely. Pulling the API anyway buys a check. The two files are joined on GEOID20, the block's fifteen-character identifier, and every one of the 2,368,443 identifiers from the API matched a shape in TIGER, in all 52 states. A join that lands at 100% is evidence the key was read correctly; a join that lands at 97% usually means an identifier was read as a number somewhere, because a fifteen-digit block id becomes `1.3121e+14` and a leading zero disappears silently.

That is the single most common way this work goes wrong. Every identifier here is held as text.

02 How the map was made

The pipeline is four steps, and each one throws something away on purpose.

1. **Ask the API** for P1_001N for every block, county by county, and keep the identifiers where the count is zero.
2. **Join** that list to the TIGER shapes for each state, keeping only the matching blocks. Per state, so that a single state's blocks are in memory at a time rather than the whole country.
3. **Dissolve** the surviving blocks into one shape. Touching empty blocks become one region, which is honest here: the boundary between two adjacent empty blocks is not a feature of the world, it is an artifact of where the Bureau put its lines.
4. **Simplify** with mapshaper, project to Albers USA, and round the coordinates to whole units of the figure's own grid.

Simplification is the step that deserves suspicion, so here are its terms. The national layer at full resolution is about 200 MB, far too large to put in a web page. What ships is 2% of the original vertices: 321,181 points in 18,200 polygons, about 1 MB. At the width this figure is drawn, one unit of the grid is roughly 2.3 kilometers, so detail finer than that could not be seen even if it were kept. What is lost is the exact shape of any individual block. What survives is where the empty land is, which is the only claim the figure makes.

Two things in step 3 and 4 are easy to get wrong and were got wrong here first. Dissolving produces polygons with **holes** – a populated town inside an empty county is an interior ring – and a ring drawn on its own is painted as fill rather than subtracted, which turns Nevada into a solid rectangle. Both renderings below therefore draw one path per polygon with all of its rings together, under an even-odd fill rule. And rounding coordinates has to happen *after* projecting, not before: rounding longitude and latitude to whole numbers rounds them to whole degrees, and the country dissolves.

The built tables are small and readable: `state_blocks.csv`, one row per state; `block_pop_hist.csv`, a count of blocks at each population from 0 to 1,000; `zero_land.csv`, land area of the empty blocks by state; `facts.csv`, the national totals quoted in this text; and the figure geometry in `map_zero.csv`, `map_states.csv` and `map_frame.csv`, stored as integer figure coordinates so the web version and the print version draw the same shapes from the same numbers.

Level	Where	Blocks	Blocks with nobody	Percent
Nation	50 states, DC, Puerto Rico	8,174,955	2,368,443	29.0%
Most empty	Alaska	28,568	16,803	58.8%
Least empty	Puerto Rico	41,987	5,417	12.9%

03 What the map shows

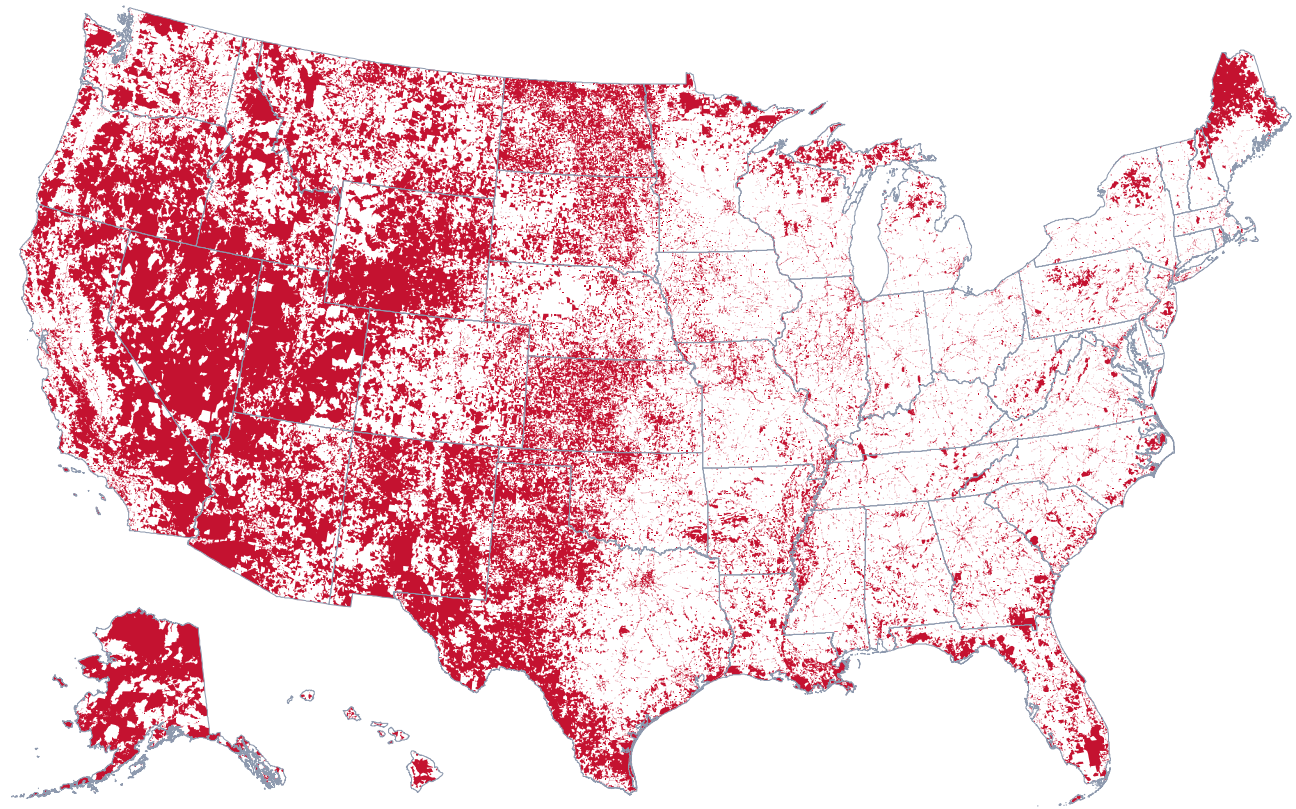


Figure 1. Every census block in the United States that recorded no residents in 2020.

Red is empty, white is inhabited land, and the map draws the 2,130,094 empty blocks that have land area. A single flat color fits the question, because the claim is about where, not how much.

Read west to east and the map turns into a history of land ownership. The solid masses through Nevada, Idaho and eastern Oregon are federal land: national forest, Bureau of Land Management range, military reserve. The sharp vertical seam around the hundredth meridian is where homestead-scale agriculture stopped being viable. In the East the pattern inverts into a fine red lacework that traces rivers, interstates and the edges of reservoirs, because east of the Mississippi the empty blocks are mostly small and linear rather than large and areal.

The number under the map is the one that surprises people. Those 29.0% of blocks cover about 1,465,387 square miles, roughly 41.5% of the land area of the United States. **Blocks are not land.** A statistic about how many blocks are empty and a statistic about how much of the country is empty are different statistics, and the gap between 29.0% and 41.5% is the size of the difference.

04 How big is a census block, anyway?

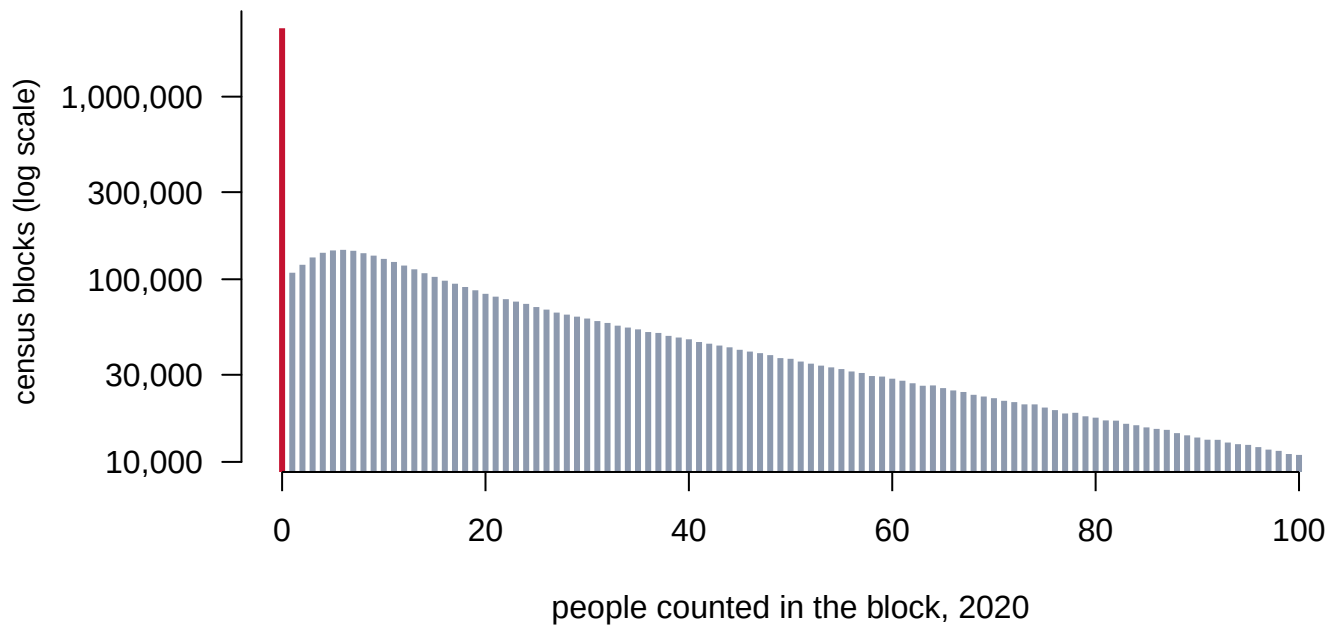


Figure 2. How many people live in a census block. One bar per exact population from 0 to 100; the red bar at the left is the 2,368,443 blocks with nobody. The vertical axis is logarithmic, because the zero bar is more than twenty times the next tallest and a linear axis would show a single spike above a flat floor.

The shape says the block is a small thing. The median block in the country holds 14 people. Among blocks with anybody at all, the median is 28. 45.3% of all blocks hold ten people or fewer. A census block is not a neighborhood; it is closer to a city block, which is what it was named for.

This matters for reading any block-level statistic. A percentage computed on a unit holding fourteen people is mostly noise, and the Bureau's disclosure-avoidance noise is deliberately largest, in relative terms, exactly here.

05 Which states are emptiest

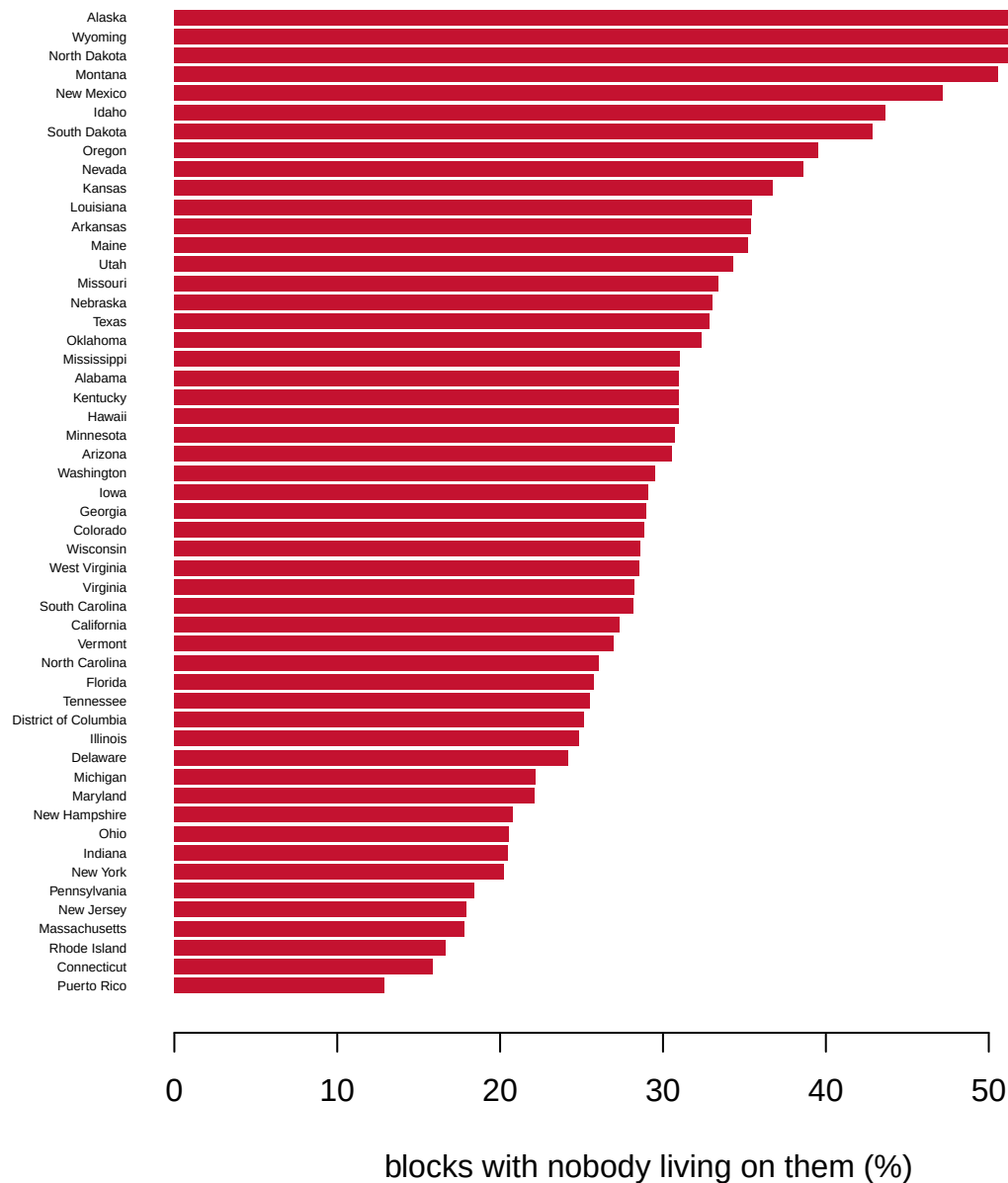


Figure 3. The share of each state's blocks that recorded nobody. Sorted rather than alphabetical, because the ordering is the finding.

The range runs from Alaska at 58.8% to Puerto Rico at 12.9%. It is tempting to read this as a map of emptiness, and it is partly that. But it is also a map of how the Bureau cut each state. Blocks are drawn along visible features, so a state with a dense grid of section-line roads gets many small blocks whether or not anyone lives on them, and a state with a lot of water gets a lot of water blocks. Alaska is genuinely empty; it also has an enormous coastline and river network, each stretch of which is its own block.

06 What this brief cannot tell you

The 29.0% is a fact about blocks, not about people or land, and the three do not track each other. Nothing here says how much of the country is *uninhabitable*, only which units recorded no usual residents on 1 April 2020. A block containing a large prison, a dormitory closed for the pandemic, or a seasonal resort counted empty that day may be busy most of the year.

The map is drawn from the land blocks. 238,349 of the empty blocks are water and are not drawn, because including them fills the Great Lakes and every coastal bay with solid color and makes the map a map of hydrography. That is a presentation choice, and it is reversible; the land-area figures are unaffected either way, because the Bureau's ALAND20 already excludes water.

Simplification means no individual block on this map is the right shape. Do not measure anything off it. The built tables carry the counts; the figure carries the pattern.

And zero is a published number, not a raw one. The 2020 counts passed through disclosure avoidance before release, which can move a small count by a little. At the block level that noise is proportionally largest, though a block that is a reservoir was not going to be populated either way.

07 What you should have learned

- The census block is small: the median holds 14 people, and 45.3% hold ten or fewer. Percentages computed on units that size are fragile.
- 29.0% of blocks are empty but they cover about 41.5% of the land. Counting units and measuring area are different operations, and “most of the country is empty” and “most blocks are empty” are different claims.
- A join that matches 100% is a test you passed, not a formality. Identifiers are text.
- Every step that made this map — filtering, dissolving, simplifying — discarded something. A figure is trustworthy when you can say what it discarded and why, not when nothing was discarded.

08 Extensions

- `state_blocks.csv` — divide pop by blocks for each state. Which states put the fewest people in a block, and is that the same list as the emptiest states in Figure 3?
- `state_blocks.csv` joined to `zero_land.csv` — rank states by `pct_zero`, then by `zero_land_sqmi`. Name a state that is high on one list and low on the other, and say what the Bureau did to produce that gap.
- `block_pop_hist.csv` — the last row is a bucket, every block of 1,000 people or more. How many blocks are in it, and what kind of place holds that many people inside one block?
- `block_pop_hist.csv` — compute the share of the national population living in blocks of ten people or fewer. Compare it with the 45.3% of *blocks* that size. Explain the difference to someone in one sentence.
- **Stretch.** The TIGER block file carries POP20 already. Download one state's `tabblock20` and check its POP20 against the API's `P1_001N` for the same GEOIDs. Report how many

disagree, and decide which you would trust in print.

- **Stretch.** Rebuild Figure 1 for one state without the dissolve step, drawing each empty block separately. Say what the dissolved version hides and what it clarifies.

09 Sources

Data

- U.S. Census Bureau, 2020 Census Redistricting Data (P.L. 94-171) Summary File, served through the Census Data API as the 2020/dec/pl dataset; variable P1_001N is total population, and summary level block is the Bureau's smallest published unit. <https://api.census.gov/data/2020/dec/pl> and <https://www.census.gov/data/developers/data-sets/decennial-census.html>
- U.S. Census Bureau, TIGER/Line Shapefiles 2020, TABBLOCK20 layer — the boundary of every 2020 tabulation block, one file per state, carrying GEOID20, ALAND20 and AWATER20. <https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/>
- U.S. Census Bureau, Cartographic Boundary Files 2020, state layer at 1:500,000 — generalized outlines drawn for mapping rather than measurement, clipped to the shoreline, used here only as the frame of Figure 1. <https://www.census.gov/geographies/mapping-files/time-series/geo/cartographic-boundary.html>

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`block_pop_hist.csv`, `facts.csv`, `map_frame.csv`, `map_states.csv`, `map_zero.csv`, `state_blocks.csv`, `zero_land.csv`

Method

- Bloch, M. *mapshaper* — the simplification, dissolve and projection steps in Figure 1 are all mapshaper commands; its Visvalingam implementation is what “2% of the vertices” means here. <https://github.com/mbloch/mapshaper>
- Bostock, M. *D3* — the three figures in the HTML edition are plain SVG built with D3 scales and axes. <https://d3js.org/>

Documentation

- U.S. Census Bureau, “Glossary: Census Block” — the definition, and the rule that blocks are bounded by visible features and by legal boundaries. <https://www.census.gov/programs-surveys/geography/about/glossary.html>
- U.S. Census Bureau, “Disclosure Avoidance and the 2020 Census” — why a published block count is not a raw block count. <https://www.census.gov/programs-surveys/decennial-census/decade/2020/planning-management/process/disclosure-avoidance.html>

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any

starting files. Please do the following and show your work.

THE FIRST SOURCE. The U.S. Census Bureau's Data API, 2020 Decennial Census

P.L. 94-171 Redistricting Summary File:

<https://api.census.gov/data/2020/dec/pl>

Free, but it needs a key from

https://api.census.gov/data/key_signup.html.

The variable for total population is P1_001N. The API will not return every

block in a state at once: a block query has to name a state AND a county, so

covering the country takes about 3,100 requests. Block GEOIDs are 15-character strings – read them as text, never as numbers, or a 15-digit

identifier becomes 1.3121e+14 and leading zeros vanish.

THE SECOND SOURCE. TIGER/Line 2020 tabulation block shapes, one zipped shapefile per state:

https://www2.census.gov/geo/tiger/TIGER2020/TABBLOCK20/tl_2020_<FIPS>_tabblock20.zip

About 10 GB for the whole country. The attribute table carries GEOID20, ALAND20 (land area, square meters) and AWATER20.

WHAT TO BUILD.

1. A table of every census block in the 50 states, DC and Puerto Rico with

its 2020 population. Report the national total, and how many blocks recorded zero.

2. A count of blocks at each exact population from 0 to 1,000, with everything

above 1,000 in one bucket. Report the median block population, and the median among blocks with at least one resident.

3. A per-state table: blocks, blocks with nobody, and the percentage.

4. A national map of the zero-population blocks. Join the zero GEOIDs to the

TIGER shapes state by state, keep the matches, and dissolve touching blocks together. Report what fraction of the zero blocks have land area (the rest are water) and how many square miles the land ones cover.

THINGS THAT WILL BITE YOU.

- Check the join. Every GEOID from the API should match a TIGER shape. If it does not, you read an identifier as a number somewhere.
- Water blocks have zero population and are real. Decide whether they belong on your map and say which you chose; drawing them fills the Great Lakes.
- Dissolving makes polygons with holes. Draw each polygon's rings together under an even-odd fill rule, or the holes get painted as fill.
- Round coordinates only AFTER projecting. Rounding longitude and latitude rounds to whole degrees.

CHECK YOUR WORK. The copies this book used were downloaded in September 2026. If your rebuild matches them, you should find:

- 8,174,955 blocks holding 334,735,155 people, across 52 states and state-equivalents
- 2,368,443 blocks with nobody living on them – 29.0%
- a median block population of 14, or 28 among blocks with at least one resident; 45.3% of blocks hold ten people or fewer
- emptiest Alaska at 58.82%, least empty Puerto Rico at 12.90%
- 2,130,094 of the empty blocks have land area, covering 1,465,387 square miles – about 41% of U.S. land, against 29% of blocks
- every one of the 2,368,443 API identifiers matches a TIGER shape; if yours does not, you read a GEOID as a number

Both sources are finished 2020 census products; they will not change until the 2030 cycle publishes successors.